SCREEN SENSE – Kids' Screentime Visualization

Consolidated Internship Report (Weeks 1–6)

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1. Project Overview

The *ScreenSense: Kids' Screentime Visualization* project focuses on analysing screentime habits of Indian children using data cleaning, feature engineering, visual exploration, and dependency modelling.
The insights help parents, educators, and policymakers understand digital behaviour patterns and potential health risks.

Dataset used: **Indian Kids Screentime 2025 (Kaggle)**

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2. Milestone 1 — Data Foundation & Cleaning

Week 1 – Dataset Understanding

Key Actions:

- Loaded and verified the dataset (9,712 rows × 8 columns).
- Explored schema, data types, summary statistics.
- Checked unique values for each column.
- Created initial visualizations:
  - Distribution of Screen Time
  - Gender Distribution
  - Urban vs Rural Distribution

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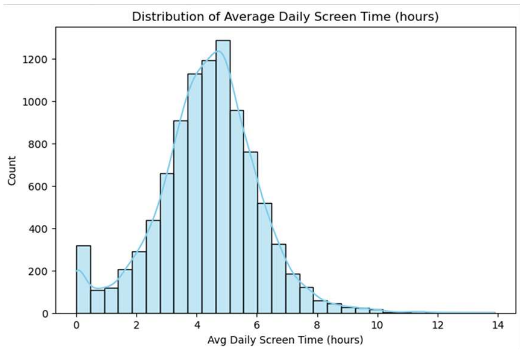
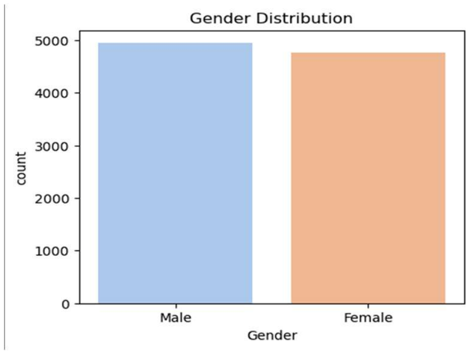
 Dataset Information:

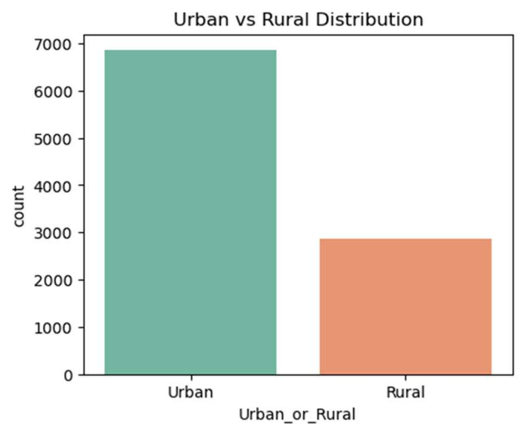
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9712 entries, 0 to 9711
Data columns (total 8 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Age                               9712 non-null   int64
 1   Gender                            9712 non-null   object
 2   Avg_Daily_Screen_Time_hr          9712 non-null   float64
 3   Primary_Device                    9712 non-null   object
 4   Exceeded_Recommended_Limit        9712 non-null   bool
 5   Educational_to_Recreational_Ratio 9712 non-null   float64
 6   Health_Impacts                    6494 non-null   object
 7   Urban_or_Rural                    9712 non-null   object
dtypes: bool(1), float64(2), int64(1), object(4)
memory usage: 540.7+ KB
```

Key Insights:

- Majority of kids spend **3–6 hours daily**.
- Smartphones dominate usage (~45%).
- Dataset is clean with no missing values initially.







Week 2 – Preprocessing & Feature Engineering

Data Cleaning Performed:

- Removed whitespace, standardized text.
- Fixed inconsistent categories:
  - 'male' → 'Male', 'urban' → 'Urban'.
- Filled missing entries in *Primary_Device* & *Health_Impacts*.

Feature Engineering:
Created derived fields including:

- **Age Band**
- **Screen Time Level** (Low / Moderate / High)
- **Health Impact Count**
- **Urban/Rural Flag**

- **Overuse Index & Risk Category**
- **Usage Behaviour Type**
- **Weekend/Weekday Usage Indicator**

These enriched features enabled deeper visual analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Age	Gender	Avg_Daily_Screen_Time_hr	Primary_Device	Exceeded_Recommended_Limit	Educational_to_Recreational_Ratio	Health_Impacts	Urban_or_Rural	Age_Band	ScreenTime_Level	Edu_Recreational_%	Health_Impact_Count	Urban_Rural_Flag
2	14	Male	3.99	Smartphone	TRUE	0.42	Poor Sleep, Eye Strain	Urban	Teen (14-16)	Moderate (3-6 hr)	42	2	1
3	11	Female	4.61	Laptop	TRUE	0.3	Poor Sleep	Urban	Pre-Teen (11-13)	Moderate (3-6 hr)	30	1	1
4	18	Female	3.73	Tv	TRUE	0.32	Poor Sleep	Urban	Young Adult (17-18)	Moderate (3-6 hr)	32	1	1
5	15	Female	1.21	Laptop	FALSE	0.39	None	Urban	Teen (14-16)	Low (<3 hr)	39	0	1
6	12	Female	5.89	Smartphone	TRUE	0.49	Poor Sleep, Anxiety	Urban	Pre-Teen (11-13)	Moderate (3-6 hr)	49	2	1
7	14	Female	4.88	Smartphone	TRUE	0.44	Poor Sleep	Urban	Teen (14-16)	Moderate (3-6 hr)	44	1	1
8	17	Male	2.97	Tv	FALSE	0.48	None	Rural	Young Adult (17-18)	Low (<3 hr)	48	0	0
9	10	Male	2.74	Tv	TRUE	0.54	None	Urban	Child (8-10)	Low (<3 hr)	54	0	1
10	14	Male	4.61	Laptop	TRUE	0.36	Poor Sleep, Anxiety	Rural	Teen (14-16)	Moderate (3-6 hr)	36	2	0
11	18	Male	3.24	Tablet	TRUE	0.48	Poor Sleep, Obesity Risk	Urban	Young Adult (17-18)	Moderate (3-6 hr)	48	2	1
12	18	Male	3.53	Tablet	TRUE	0.44	Poor Sleep	Rural	Young Adult (17-18)	Moderate (3-6 hr)	44	1	0
13	15	Female	4.94	Tv	TRUE	0.36	Poor Sleep	Rural	Teen (14-16)	Moderate (3-6 hr)	36	1	0
14	12	Male	4.59	Smartphone	TRUE	0.43	Poor Sleep	Urban	Pre-Teen (11-13)	Moderate (3-6 hr)	43	1	1
15	11	Male	6.08	Tv	TRUE	0.48	Eye Strain	Rural	Pre-Teen (11-13)	High (>6 hr)	48	1	0
16	15	Male	6.15	Smartphone	TRUE	0.4	Poor Sleep	Urban	Teen (14-16)	High (>6 hr)	40	1	1
17	15	Female	3.69	Smartphone	TRUE	0.34	Poor Sleep	Urban	Teen (14-16)	Moderate (3-6 hr)	34	1	1
18	10	Female	7.1	Smartphone	TRUE	0.43	Poor Sleep, Eye Strain	Rural	Child (8-10)	High (>6 hr)	43	2	0
19	13	Female	6.98	Smartphone	TRUE	0.48	Obesity Risk	Urban	Pre-Teen (11-13)	High (>6 hr)	48	1	1
20	12	Male	3.82	Smartphone	TRUE	0.4	Anxiety	Urban	Pre-Teen (11-13)	Moderate (3-6 hr)	40	1	1
21	9	Female	0	Tv	FALSE	0.42	None	Urban	Child (8-10)	Low (<3 hr)	42	0	1
22	15	Male	3.85	Smartphone	TRUE	0.49	Obesity Risk	Urban	Teen (14-16)	Moderate (3-6 hr)	49	1	1
23	13	Female	4.05	Smartphone	TRUE	0.33	Eye Strain	Urban	Pre-Teen (11-13)	Moderate (3-6 hr)	33	1	1
24	9	Female	1.63	Tablet	FALSE	0.58	None	Urban	Child (8-10)	Low (<3 hr)	58	0	1
25	12	Female	6.78	Tv	TRUE	0.33	Eye Strain	Urban	Pre-Teen (11-13)	High (>6 hr)	33	1	1
26	8	Male	3.19	Tablet	TRUE	0.54	Poor Sleep, Anxiety, Obesity Risk	Urban	Child (8-10)	Moderate (3-6 hr)	54	3	1
27	17	Male	4.96	Smartphone	TRUE	0.32	None	Urban	Young Adult (17-18)	Moderate (3-6 hr)	32	0	1
28	13	Female	1.85	Smartphone	FALSE	0.49	None	Urban	Pre-Teen (11-13)	Low (<3 hr)	49	0	1
29	16	Male	4.22	Smartphone	TRUE	0.5	Poor Sleep	Urban	Teen (14-16)	Moderate (3-6 hr)	50	1	1

		K	L	M	N	O	P	Q	R	S	T
1	Level	Edu_Recreational_%	Health_Impact_Count	Urban_Rural_Flag	Overuse_Index	Risk_Category	Primary_Use_Category	Device_Usage_Context	Weekday_Weekend_Usage	Usage_Behavior_Type	Health_Risk_Text
2	6 hr)	42	2	1	2.31	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
3	6 hr)	30	1	1	3.23	Medium Risk	Mixed Use	Urban Laptop	Higher Usage on Weekends	Recreational Focused	Moderate Risk
4	6 hr)	32	1	1	2.54	Low Risk	Mixed Use	Urban Tv	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
5		39	0	1	0.74	Low Risk	Mixed Use	Urban Laptop	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
6	6 hr)	49	2	1	3	Low Risk	Mixed Use	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage
7	6 hr)	44	1	1	2.73	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage
8		48	0	0	1.54	Low Risk	Mixed Use	Rural Tv	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
9		54	0	1	1.26	Low Risk	Mixed Use	Urban Tv	Lower Usage on Weekdays	Study Focused	Healthy Usage
10	6 hr)	36	2	0	2.95	Low Risk	Mixed Use	Rural Laptop	Higher Usage on Weekends	Recreational Focused	Healthy Usage
11	6 hr)	48	2	1	1.68	Low Risk	Gaming / Casual	Urban Tablet	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
12	6 hr)	44	1	0	1.98	Low Risk	Gaming / Casual	Rural Tablet	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
13	6 hr)	36	1	0	3.16	Medium Risk	Mixed Use	Rural Tv	Higher Usage on Weekends	Recreational Focused	Moderate Risk
14	6 hr)	43	1	1	2.62	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage
15		48	1	0	3.16	Medium Risk	Mixed Use	Rural Tv	Higher Usage on Weekends	Heavy but Balanced User	Moderate Risk
16		40	1	1	3.69	Medium Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Higher Usage on Weekends	Heavy but Balanced User	Moderate Risk
17	6 hr)	34	1	1	2.44	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
18		43	2	0	4.05	Medium Risk	Entertainment (YouTube/OTT)	Rural Smartphone	Higher Usage on Weekends	Heavy but Balanced User	Moderate Risk
19		48	1	1	3.63	Medium Risk	Mixed Use	Urban Smartphone	Higher Usage on Weekends	Heavy but Balanced User	Moderate Risk
20	6 hr)	40	1	1	2.29	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
21		42	0	1	0	Low Risk	Mixed Use	Urban Tv	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
22	6 hr)	49	1	1	1.96	Low Risk	Mixed Use	Urban Smartphone	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
23	6 hr)	33	1	1	2.71	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage
24		58	0	1	0.68	Low Risk	Study / Educational	Urban Tablet	Lower Usage on Weekdays	Study Focused	Healthy Usage
25		33	1	1	4.54	Medium Risk	Mixed Use	Urban Tv	Higher Usage on Weekends	Overuse - Recreational Heavy	Moderate Risk
26	6 hr)	54	3	1	1.47	Low Risk	Gaming / Casual	Urban Tablet	Lower Usage on Weekdays	Balanced User	Healthy Usage
27	6 hr)	32	0	1	3.37	Low Risk	Entertainment (YouTube/OTT)	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage
28		49	0	1	0.94	Low Risk	Mixed Use	Urban Smartphone	Lower Usage on Weekdays	Recreational Focused	Healthy Usage
29	6 hr)	50	1	1	2.11	Low Risk	Mixed Use	Urban Smartphone	Higher Usage on Weekends	Balanced User	Healthy Usage
30		51	0	1	0.15	Low Risk	Mixed Use	Urban Smartphone	Lower Usage on Weekdays	Study Focused	Healthy Usage
31	6 hr)	43	2	1	2.92	Low Risk	Mixed Use	Urban Tv	Higher Usage on Weekends	Recreational Focused	Healthy Usage
32	6 hr)	48	1	1	2.31	Low Risk	Mixed Use	Urban Smartphone	Higher Usage on Weekends	Recreational Focused	Healthy Usage

3. Milestone 2 — Visual Exploration & Trends

Week 3 – Univariate & Bivariate Visual Analysis

Univariate Visuals Included:

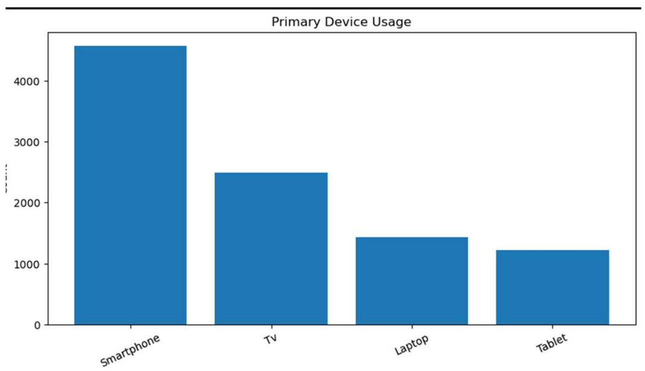
- Age Band Distribution
- Device Preference
- Health Issues by Device
- Distribution of Screen Time Levels

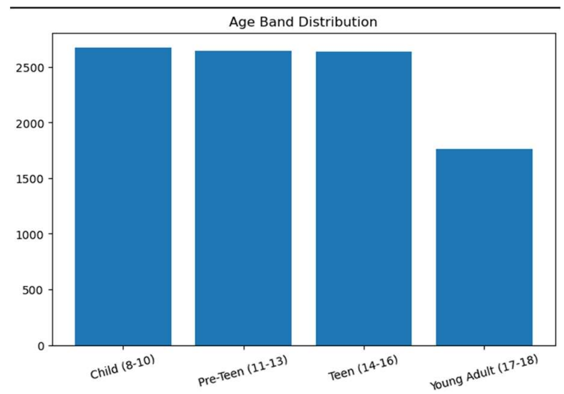
Bivariate Visuals Included:

- Screen Time by Age Band
- Screen Time by Gender
- Device Preference by Gender
- Educational vs Recreational Ratio

Key Findings:

- Smartphones remain the most used device.
- Teenagers show **higher screen exposure**.
- Males have slightly more high-usage outliers.
- Educational vs recreational ratio decreases with age.

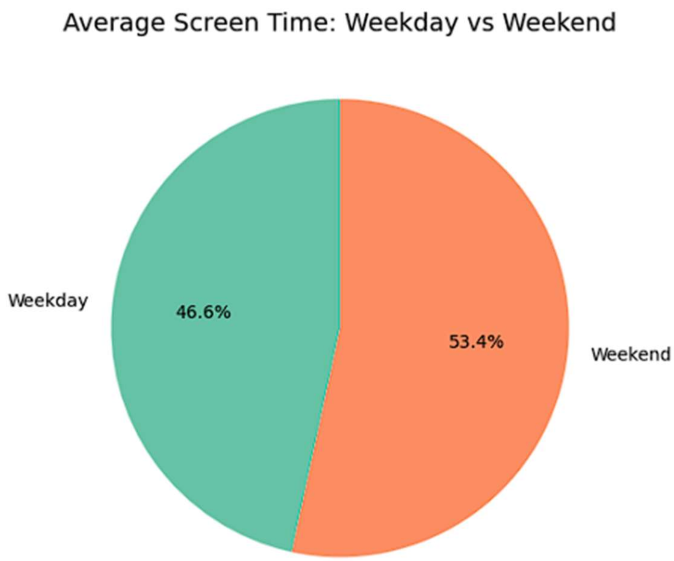


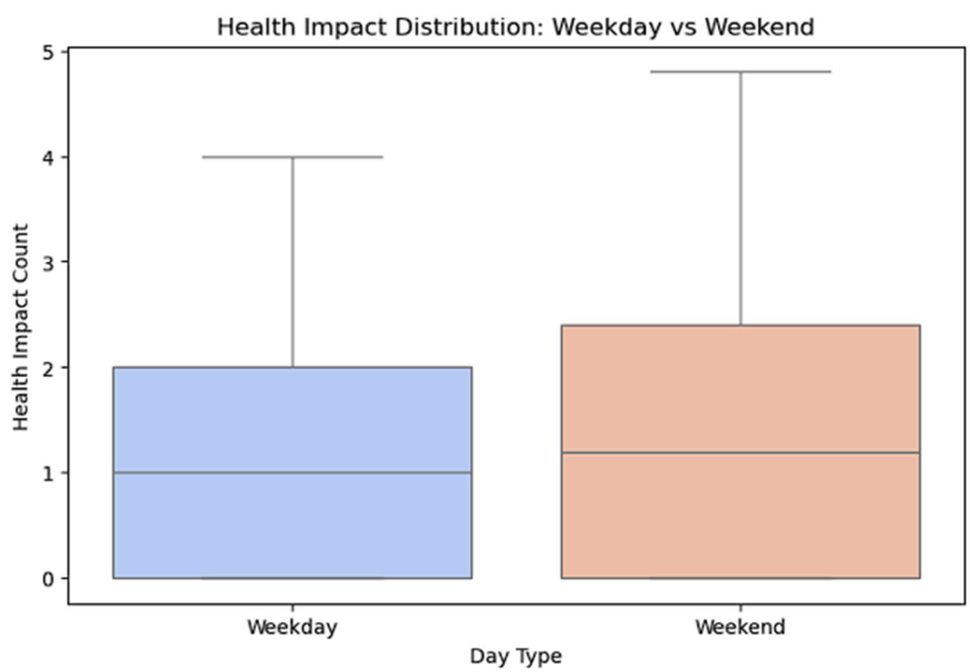


Week 4 – Weekday vs Weekend Behaviour

Insights:

- Weekend screen time is significantly higher.
- More kids fall into **High Screen-Time (>6 hrs)** on weekends.
- Health impacts increase on weekends.





4. Milestone 3 — Cohort & Dependency Insights

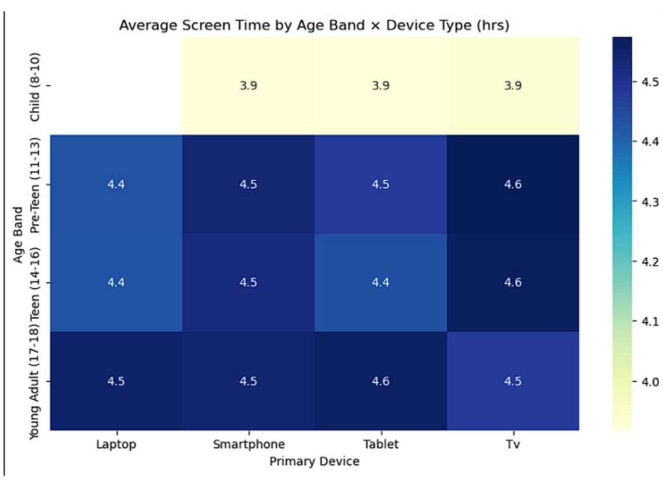
Week 5 – Cohort & Segment Analysis

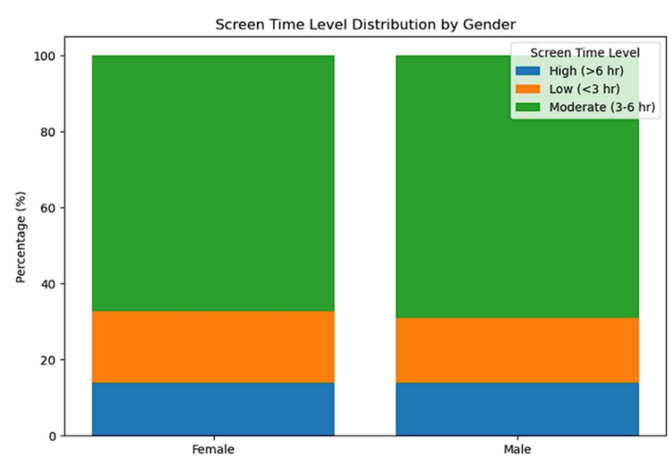
Advanced Visuals Built:

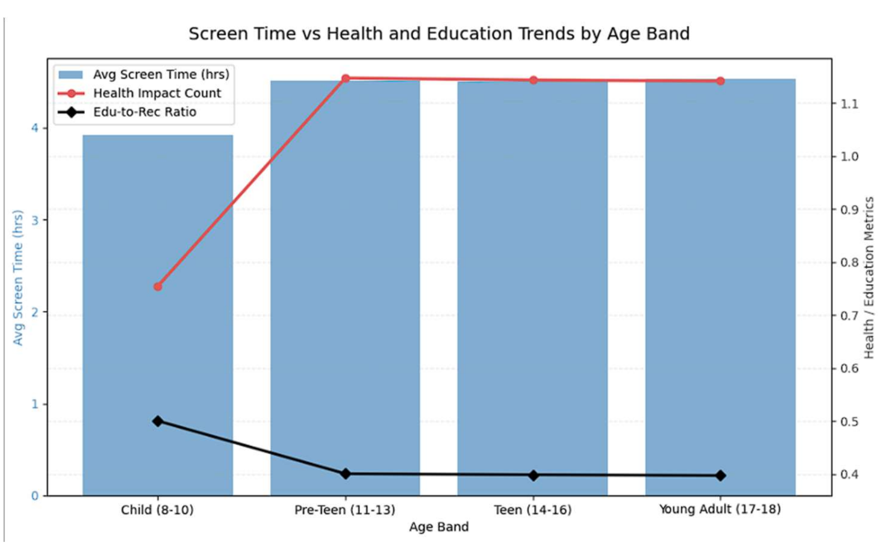
- Heatmap: Age Band × Device Type.
- Gender-wise Screen Time Levels
- Device Type × Gender (strip plot)
- Ridgeline Plot: Screen Time Distribution by Age Group
- Health Impacts by Region & Age Band

Key Insights:

- Older children use smartphones & laptops more.
- Boys show Slightly more high-usage cases.
- Urban kids report slightly higher digital-health issues
- Screen time and health impacts increase consistently with age.







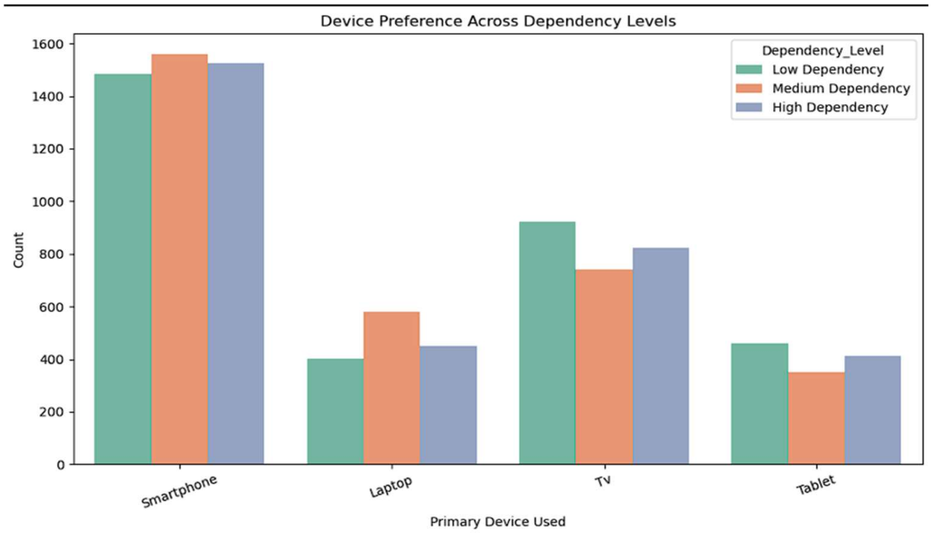
Week 6 – Digital Dependency Scoring

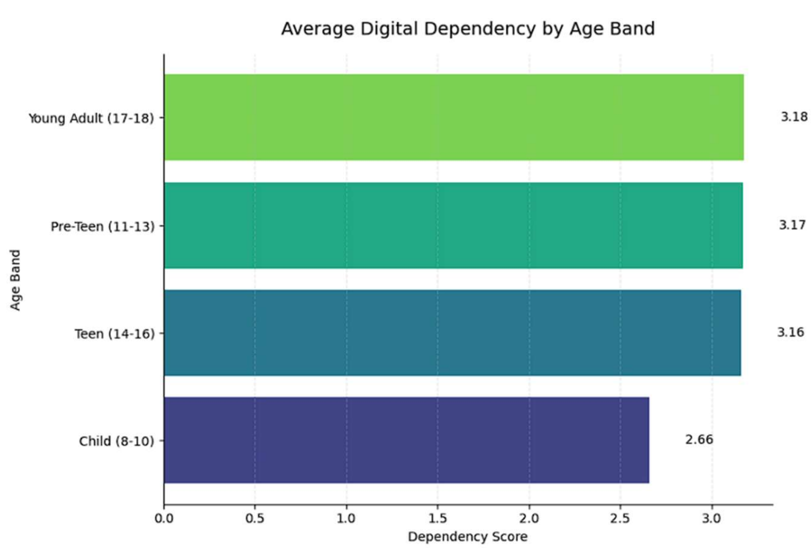
What Was Done:

- Created a **Digital Dependency Score**.
- Segmented users into:
    - Low Dependency
    - Medium Dependency
    - High Dependency

Findings:

- High-dependency children prefer smartphones heavily.
- Urban kids have slightly higher dependency due to easier access.
- Dependency increases sharply with age.





5. Final Project Summary

Across Weeks 1–6, the project progressed from raw dataset exploration to advanced behavioural and dependency insights.

Major Conclusions:

- Screen-time increases steadily with age.
- Smartphones are the dominant device across all demographics.
- Weekends show **higher overuse** and **higher health impacts**.
- Urban children are more exposed to screens and face more digital-health challenges.
- Teenagers exhibit the **highest dependency and risk levels**.
- Recreational usage increases, while educational ratio declines with age.

These insights support improved parental guidance, policy recommendations, and screen-time balance strategies.